Tebako Specifications — Series landing

:2026, Version 1.0

The Tebako Project

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Tebako
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Ribose

Contents

1. The series 4

2. Source of truth and synchronization 4

3. Documents in the series 5

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The Tebako Project

Suite 1, 8/F, New Henry House
10 Ice House Street
Central
Hong Kong

copyright@ribose.com
www.ribose.com

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1. The series

The tebako ecosystem packages applications and runtimes into single executables whose loading behavior is governed by normative specifications: the wire format of the package container, the manifest grammar of a payload, the contract between the loader and a runtime, and the policies for resolution, caching, jailing, trust, and observability. Those normative texts live as Markdown files in the product repository, and an implementer — a runtime factory author, a registry operator, a feedstock maintainer — has until now read them only as raw Markdown source, with no rendered document to cite.

The Tebako Specifications (TBK SPEC) series closes that gap. Each document in the series is one normative text from the repository, converted to a Metanorma standard document and rendered to HTML and PDF as a member of a single collection. The document numbering follows the numbering of the source files: TBK SPEC 02 renders `02-tpkg-wire-format.md`, and so on. Section numbering, section titles, tables, and code listings are carried over verbatim, so a reference of the form "spec 06 §4" resolves identically against the Markdown source and against the rendered document.

2. Source of truth and synchronization

The Markdown files under `docs/spec/` in the `tamatebako/tebako` repository are the source of truth for every document in the series. This series is their rendered form; the AsciiDoc twin of a specification is derived from the Markdown and never edited independently. The Markdown sources themselves are not modified by the rendering process.

Each rendered document records the blob SHA of the Markdown file it was converted from in a comment line of the form `// source: docs/spec/NN-name.md @ <sha>`. The drift gate at `docs/spec-site/tools/sync.rb` recomputes the blob SHA of every referenced Markdown source and compares it with the recorded value. The gate exits with status 0 when every rendered twin is current, and exits 1 naming the drifted

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documents when a Markdown source has moved without its rendered twin being updated. A continuous-integration job runs the gate on every pull request that touches `docs/spec/`, so a specification edit cannot land without its rendered twin following it.

The conversion from Markdown to AsciiDoc is performed and reviewed by hand; the gate does not regenerate document content. After a specification edit has been carried into its twin and reviewed, `ruby docs/spec-site/tools/sync.rb --sync` updates the recorded SHAs to bless the new state.

3. Documents in the series

TABLE 1

Document	Title	Markdown source
TBK SPEC 01	System overview	`docs/spec/01-overview.md`
TBK SPEC 02	tpkg wire format (L0)	`docs/spec/02-tpkg-wire-format.md`
TBK SPEC 03	Payload manifest (L1)	`docs/spec/03-payload-manifest.md`
TBK SPEC 04	References and registries (L3)	`docs/spec/04-references-and-registry.md`
TBK SPEC 05	Resolution and machine cache (L3)	`docs/spec/05-resolution-and-cache.md`
TBK SPEC 06	Launcher ABI (bootstrap → runtime)	`docs/spec/06-launcher-abi.md`
TBK SPEC 07	Shims and dispatch	`docs/spec/07-shims-and-dispatch.md`
TBK SPEC 08	Jails: host-access policy	`docs/spec/08-jails.md`
TBK SPEC 09	Trust and signing (authentication)	`docs/spec/09-trust-and-signing.md`
TBK SPEC 10	Encryption (confidentiality)	`docs/spec/10-encryption.md`
TBK SPEC 11	TFS: the virtual filesystem layer	`docs/spec/11-tfs-vfs-model.md`
TBK SPEC 17	The runtime driver contract (any language)	`docs/spec/17-runtime-driver-contract.md`
TBK SPEC 25	Trace observability: the interception bus, diagnosis, and coverage	`docs/spec/25-trace-observability.md`
TBK SPEC 36	Release bundles (the runtime publish unit)	`docs/spec/36-release-bundles.md`

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Document	Title	Markdown source
TBK SPEC 37	The registry book: federation, aliases, credentials, the official seed	`docs/spec/37-registry-book.md`
TBK SPEC 38	OCI distribution (`tfs+oci:`)	`docs/spec/38-oci-distribution.md`
TBK SPEC 39	Lazy range-fetch mounting of runtime images	`docs/spec/39-lazy-range-mount.md`
TBK SPEC 40	The audit ledger: queries, the anchor, and the auditor's walk	`docs/spec/40-audit-ledger.md`
TBK SPEC 41	Registry transparency: completeness proofs for the public fabric	`docs/spec/41-registry-transparency.md`
TBK SPEC 42	Tree solving and lockfiles: one answer for a whole composition	`docs/spec/42-tree-solving.md`

Specifications whose number is absent from the table — the internal process documents and the documents outside the public format/contract set — are not part of this series rendering and remain readable in the repository.

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